

Speech-to-text on security jargon: readout

Jargon Bench · runs on 2026-10-04 (UTC) · prepared by Ky Gray

1. WHAT WE TESTED AND WHY

Field techs and customers use brand and part names (Verkada, PoE, Wiegand, NVR) that general speech-to-text often mishears, and a misheard word becomes a wrong ticket. We measured how often Deepgram and ElevenLabs get those terms right on 44 recorded field-style sentences, with and without each provider's keyterm boosting.

2. HEADLINE: JARGON TERMS HEARD CORRECTLY (RECORDED SPEECH)

Provider	Default	Boosted	Median latency (boosted)	List price / audio hr (boosted)
Deepgram (nova-3)	62.1%	89.7%	128 ms	\$0.34
ElevenLabs (scribe_v2)	79.3%	95.4%	472 ms	\$0.27

3. TOP 5 MANGLED TERMS, AND WHAT BOOSTING DID

Term	Deepgram heard	Deepgram default → boosted	ElevenLabs heard	ElevenLabs default → boosted
Avigilon	"a digilon"	0/6 → 6/6	"uh vigilant"	3/6 → 6/6
NVR	"mbr"	0/9 → 9/9	heard right	9/9 → 9/9
Aiphone	"ai"	0/3 → 0/3	"ai"	0/3 → 0/3
Axis	"access"	0/3 → 0/3	"access"	0/3 → 3/3
Brivo	"maribo"	0/3 → 3/3	"horiba"	0/3 → 3/3

Ranked by misses on recorded speech without boosting, both providers combined; ties alphabetical. Counts are correct occurrences across 3 repeats.

4. LATENCY AND COST

Deepgram: median 119 ms default, 128 ms boosted; \$0.26 → \$0.34 per audio hour at list price. ElevenLabs: median 451 ms default, 472 ms boosted; \$0.22 → \$0.27 per audio hour at list price. Latency was measured from a home connection and includes upload.

5. RECOMMENDATION AND CAVEATS

- Turn on keyterm boosting for field audio. On recorded speech it raised jargon accuracy from 62.1% to 89.7% with Deepgram and from 79.3% to 95.4% with ElevenLabs.
- The cost was small: median latency changed by 9 ms and 21 ms, and list price rose from \$0.26 to \$0.34 and from \$0.22 to \$0.27 per audio hour.
- On this dataset ElevenLabs was more accurate (95.4% vs 89.7% boosted) and Deepgram answered faster (median 128 ms vs 472 ms).
- Review the boost list for terms that sound like everyday words: boosting wrote "Axis" and "supervisory" where it wasn't said (5 clip-level cases across all audio).

44 recorded clips from one speaker, each sent 3 times; 58 jargon-term occurrences. Noise is steady synthetic pink noise; real job sites are usually harder. TTS audio came from ElevenLabs voices and is excluded from the headline. Batch transcription only. Models and list prices as of 2026-10-03; runs on 2026-10-04 (UTC). Full scorecard and method: github.com/mrkygray-art/nightagent/tree/main/bench